Basic Information and Academic Performance

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Data Acquisition Technology (95)
Data Structures and Algorithms (93)
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Web application technology (93)
Cloud computing and data centers (93)
Biological and medical big data (96)

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CET-4, CET-6, IELTS

Award


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Research project



HKIE Transactions

Surgi-Diff: A Diffusion-Based Generative Framework for Postoperative Outcome Prediction and Surgical Planning in Adolescent Idiopathic Scoliosis

Journal:	*HKIE Transactions*
Manuscript ID	THIE-2026-0031
Manuscript Type:	The HKIE Outstanding Paper Award for Young Engineers/Researchers 2026
Date Submitted by the Author:	01-Apr-2026
Complete List of Authors:	Meng, Nan; The University of Hong Kong Li Ka Shing Faculty of Medicine, Department of Orthopaedics and Traumatology Chen, Zishen; Shantou University, Shantou University School of Mathematics and Computer Science Zhang, Wentao; Shantou University, Shantou University School of Mathematics and Computer Science Cheung, Jason; The University of Hong Kong Li Ka Shing Faculty of Medicine, Department of Orthopaedics and Traumatology
Keywords:	Adolescent idiopathic scoliosis, Surgical planning, Diffusion model, Postoperative outcome prediction, Deep learning
Abstract:	Preoperative surgical planning for patients with adolescent idiopathic scoliosis (AIS) is inherently complex and remains highly dependent on clinical experience. Existing approaches primarily focus on predicting scalar surgical parameters, but lack imaging-level evidence that can support surgical decision-making. To address this gap, we propose Surgi-Diff, a diffusion-based generative framework for postoperative outcome prediction and surgical planning in AIS. By integrating structural priors from preoperative spinal X-rays and surgical region masks through a Spine Alignment Encoder (SAE), Surgi-Diff generates anatomically consistent postoperative X-rays and simultaneously predicts implant-related structures. The framework was developed and evaluated on a retrospective cohort of 612 AIS patients. Experimental results showed that Surgi-Diff achieved the highest SSIM and competitive PSNR among compared methods, with qualitative results demonstrating strong agreement with ground truth (GT) postoperative morphology and implant layout. These results indicate that Surgi-Diff has potential clinical value for AIS preoperative planning, postoperative outcome prediction, and surgical decision support.

URL: https://mc.manuscriptcentral.com/thie

Title of the paper

Surgi-Diff: A Diffusion-Based Generative Framework for Postoperative Outcome Prediction and Surgical Planning in Adolescent Idiopathic Scoliosis

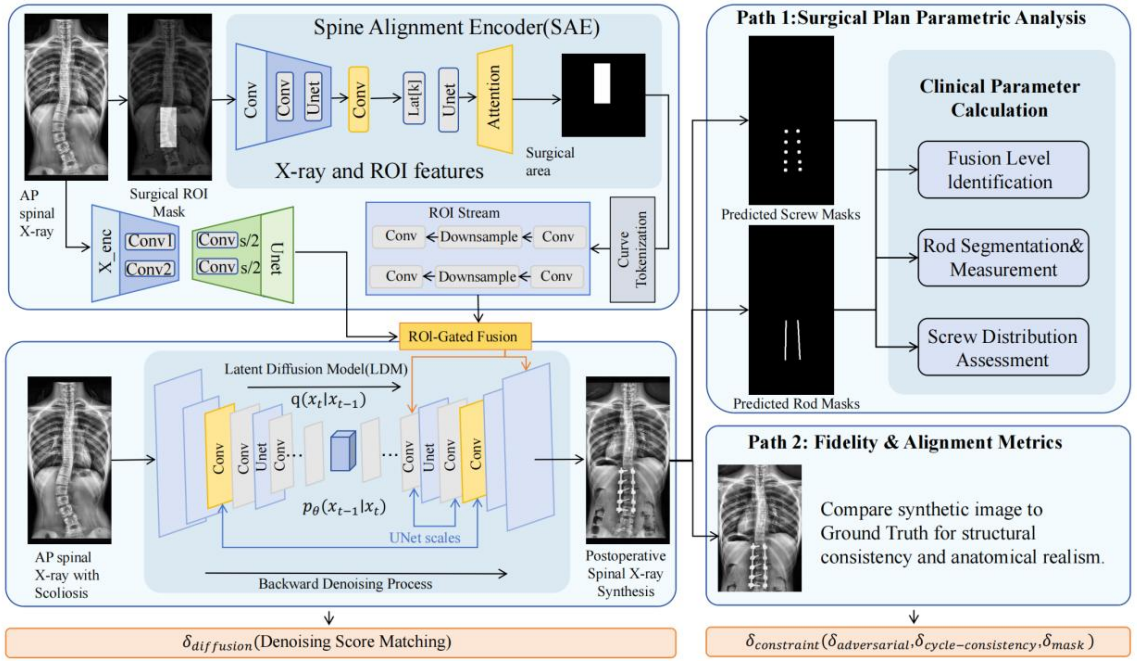
Research content

This research proposes Surgi-Diff, a diffusion generative framework integrated with a Spine Alignment Encoder, to generate anatomically consistent postoperative spinal X-rays and predict implant structures for automated preoperative surgical planning and postoperative outcome prediction of adolescent idiopathic scoliosis.

Research results

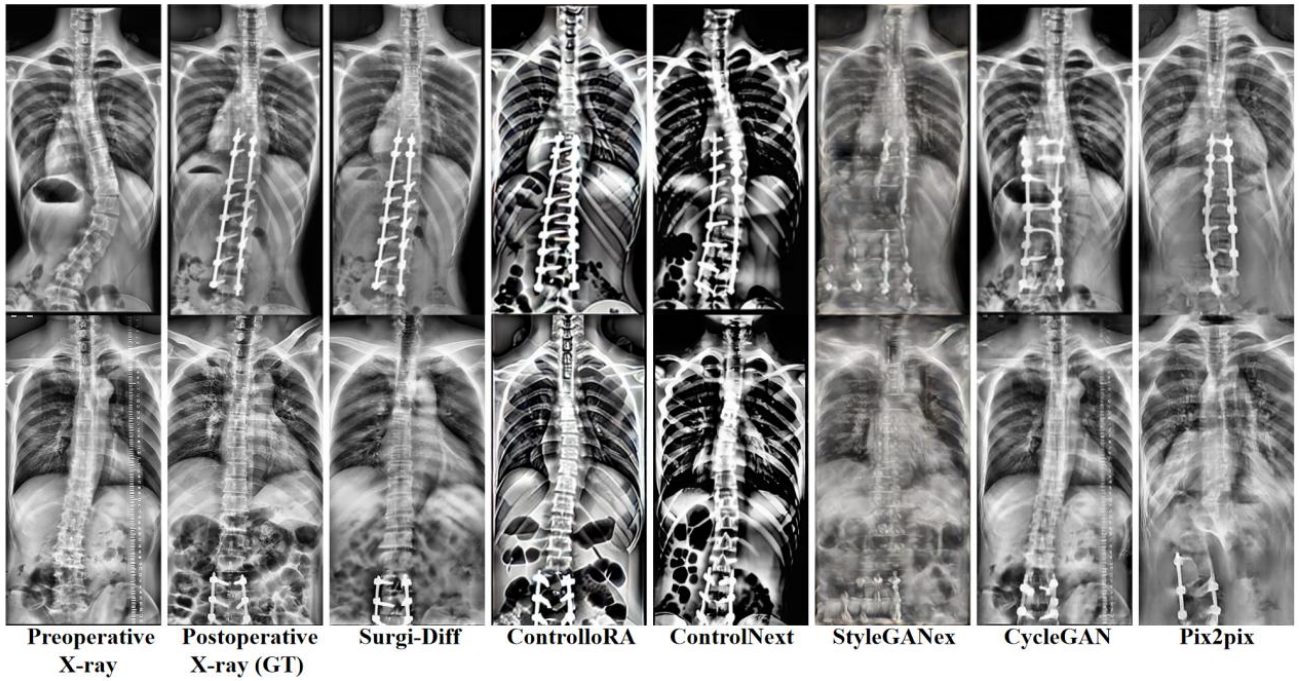
Experiments on a cohort of 612 AIS patients show that Surgi-Diff achieves the highest SSIM among all comparison methods, accurately predicts fusion segments, rods and screw distributions, and generates postoperative X-rays highly consistent with real ground truth, verifying its clinical value for scoliosis surgical decision support.

Research project



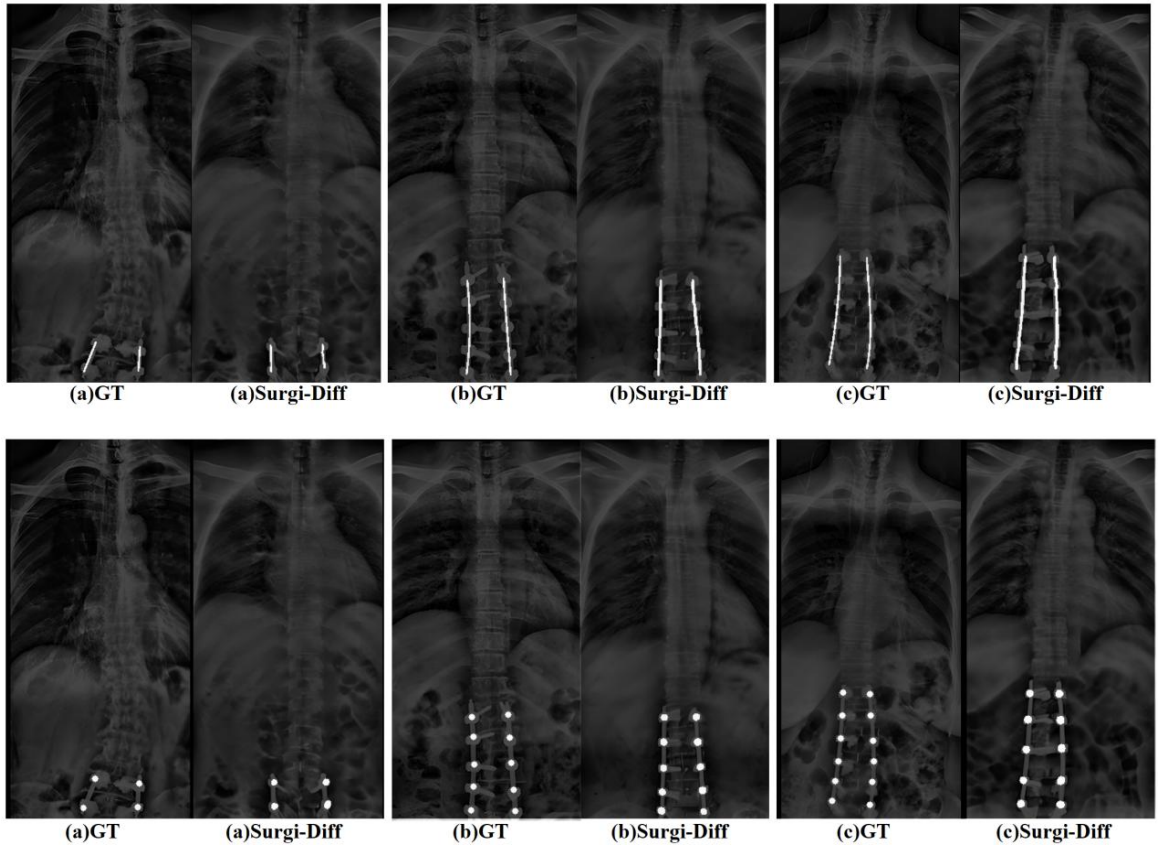
Overview of the proposed Surgi-Diff for surgical parameter prediction and postoperative spinal X-ray synthesis.

The experimental results show that Surgi-Diff achieved the highest SSIM value and competitive PSNR value in the comparison method, and the qualitative results indicate a strong consistency with the actual postoperative morphology and implant layout. These results suggest that Surgi-Diff has potential clinical value in preoperative planning and postoperative management of AIS.



method	PSNR (dB)↑		SSIM↑	
	Mean	SD	Mean	SD
Pix2pix	11.55	1.07	0.19	0.04
CycleGAN	11.49	1.12	0.20	0.05
ControlLoRA	10.05	0.88	0.14	0.03
ControlNet	8.67	0.65	0.11	0.03
StyleGANex	15.64	1.31	0.34	0.05
Surgi-Diff	14.25	2.07	0.35	0.14

Research project



Category	Metric	Symbol	Result
Fusion Level Identification	Accuracy	Acc	75.8%
	Boundary Index Error(UIV)	EUIV	8.40 pixel
	Boundary Index Error(LIV)	ELIV	7.88 pixel
Rod Geometry Measurement	Length Error	EL	19.36 pixel
	Endpoint Error	Eend	4.50 pixel
	Centerline Error	Ecline	2.41 pixel
Screw Distribution	Quantity Error	E#	1.09
	Position Error	Epos	5.49 pixel

In terms of fusion level identification, the proposed Surgi-Diff achieves an accuracy of 75.8% within a 10-pixel tolerance, with boundary index errors of 8.40 pixels for UIV (E_{UIV}) and 7.88 pixels for LIV (E_{LIV}), both below the 10-pixel threshold. For rod geometry measurement, the model delivers precise rod trajectory capture with a centerline error of only 2.41 pixels, despite a relatively large length error (EL=19.36). Regarding screw distribution, it yields an average quantity error of 1.09 and a position error of 5.49 pixels, demonstrating competent performance in localizing spinal implants. Collectively, these results verify that Surgi-Diff establishes a reliable foundation for automated spinal instrumentation analysis.

Research project



Title of the paper

G^2Net: A Geometry-Guided Network for Accurate Spine Morphology Estimation from a Single RGB Image of Human Back

Research content

This study proposes a two-stage geometry-guided G²Net for radiation-free spinal morphology estimation from single back RGB images of AIS patients. It first predicts precise back-surface depth via global-local feature fusion and Mamba modules, then leverages RGB-depth cues to train a LoRA-based conditional diffusion model to generate anatomically consistent spine curves. Experiments on a clinical dataset of 2,213 adolescents validate its superior performance against state-of-the-art methods.

Research results

Experiments on 2,213 clinical samples demonstrate that G²Net surpasses mainstream comparison methods, achieving a spine curve SSIM of 0.950 with low shape errors. Ablation tests verify the necessity of all core modules.

Research project

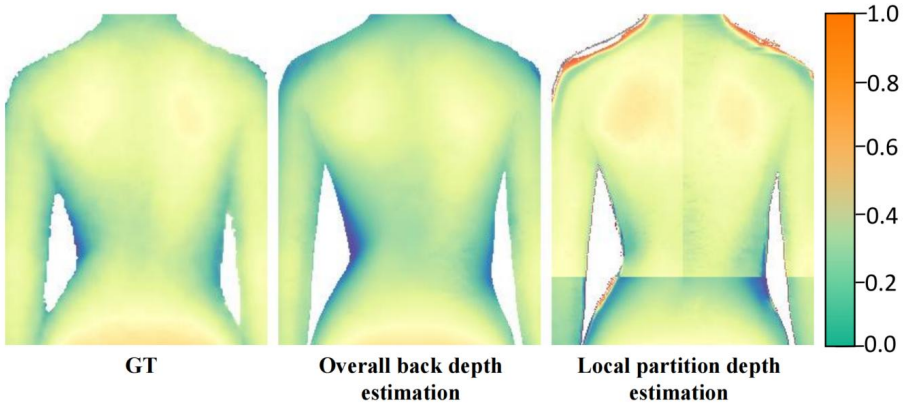
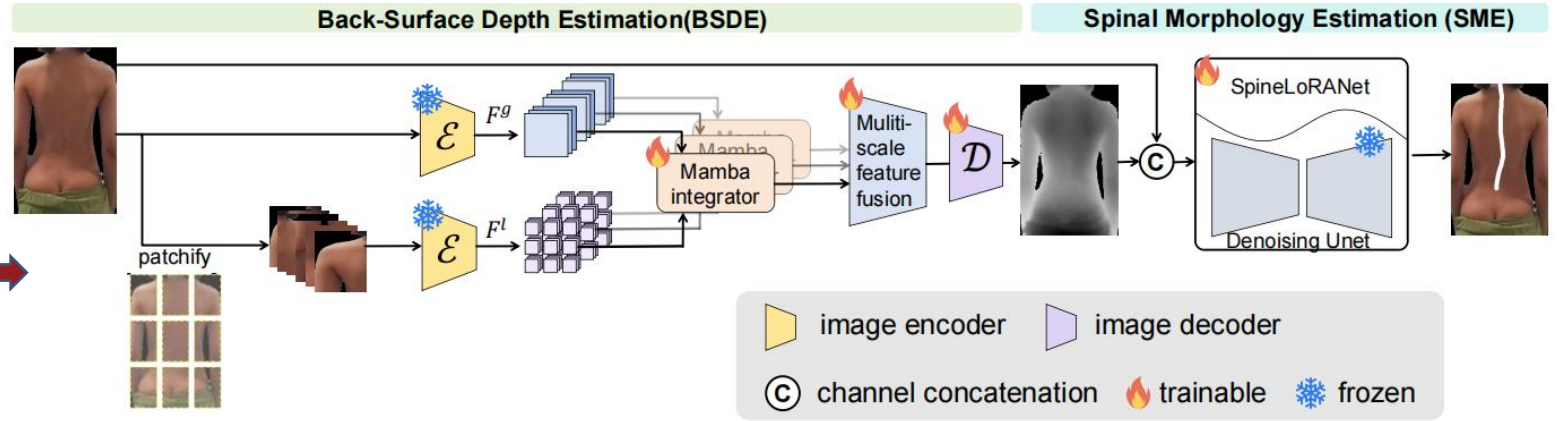


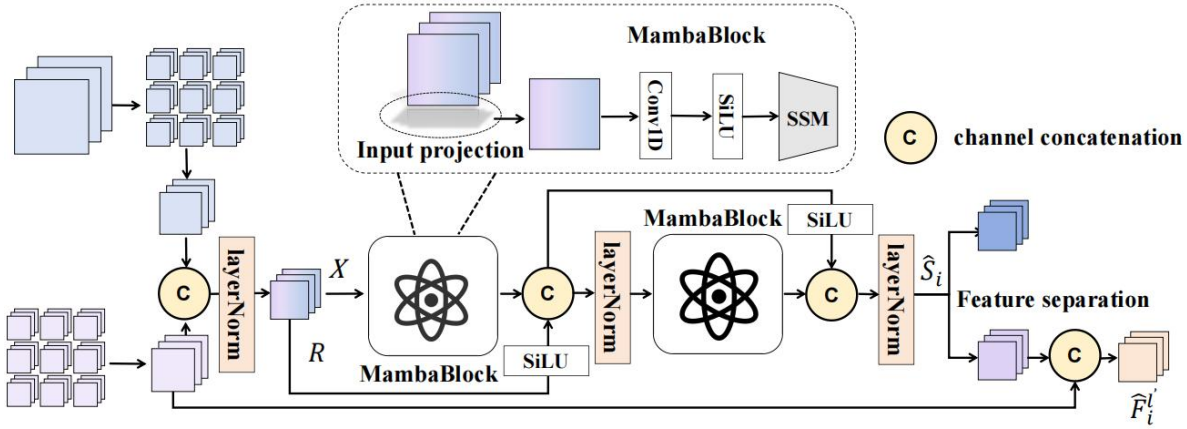
Illustration of regional inconsistency in existing monocular depth estimation methods. From left to right, real depth, overall back depth estimation and depth estimation on each local partition. These results highlight the challenge of achieving both global continuity and local detail preservation when estimating centimeter-scale back-surface depth from RGB images.

- Global depth estimation, on the other hand, can generate smoother and more consistent depth estimation maps, but at the cost of over-smoothing local fluctuations.
- Models trained based on local image blocks can achieve higher regional accuracy, but often introduce discontinuities near the boundaries of image blocks.



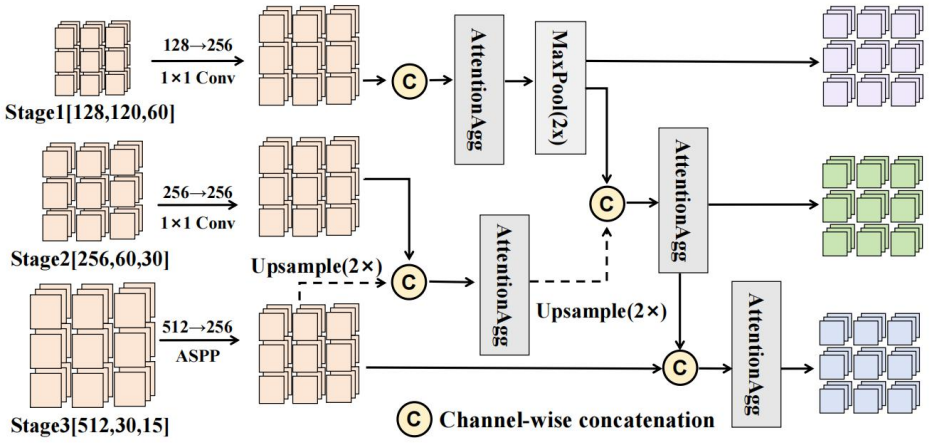
Overview of the proposed G^2Net.

To address these issues, this paper proposes G²Net, a geometric-guided framework that enables non-radiative and accurate spinal shape estimation from a single back surface RGB image. G²Net decomposes spinal shape estimation into two stages:In the first stage, high-fidelity back surface depth estimation is achieved through joint modeling of global continuity and local block-based cues. In the second stage, the estimated depth map and RGB image are used as geometric and appearance conditions to generate a compact spinal curve diagram, thereby characterizing the spinal shape.

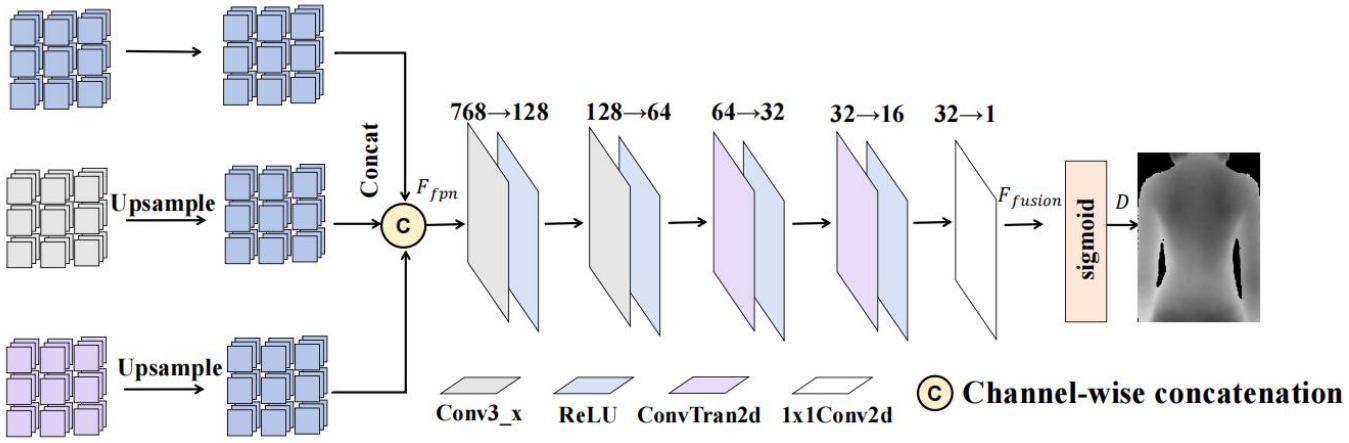


Detailed framework of the Mamba-based feature integrator, where the SSM module adopts the Structured State Space Model formulation derived from classical State Space Model theory.

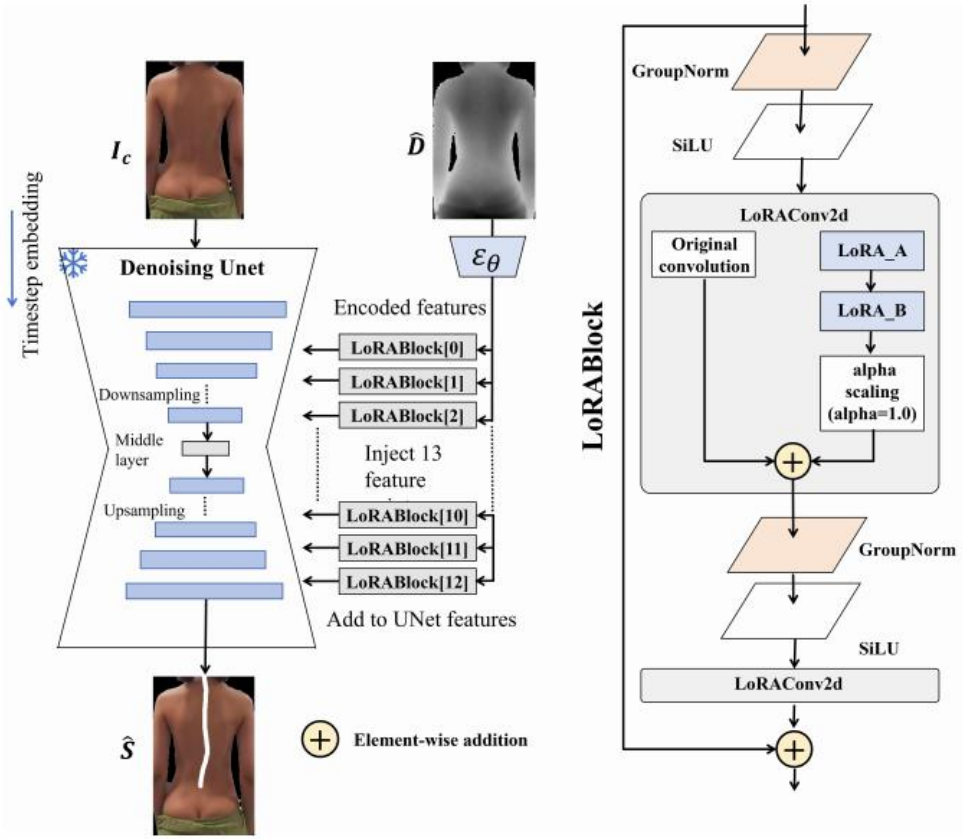
Research project



Overview of the Cross-scale Feature Fusion module, where ASPP (Atrous Spatial Pyramid Pooling) is a spatial pyramid module composed of parallel atrous convolutions with different dilation rates for multi-scale representation.

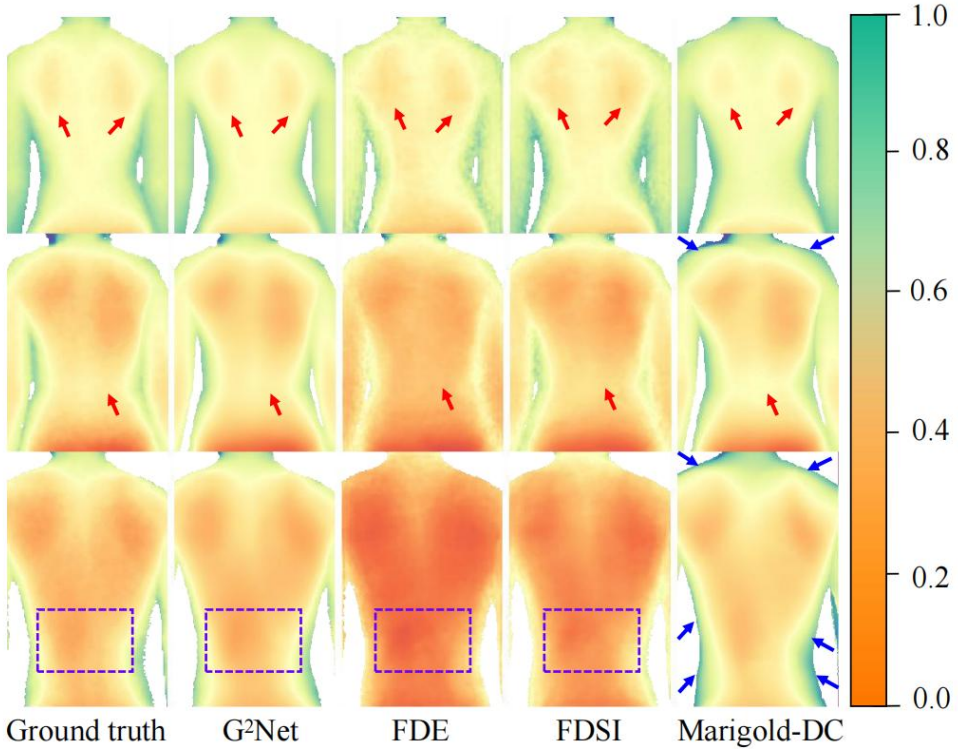


Detailed framework of the cascade upsampling decoder module.

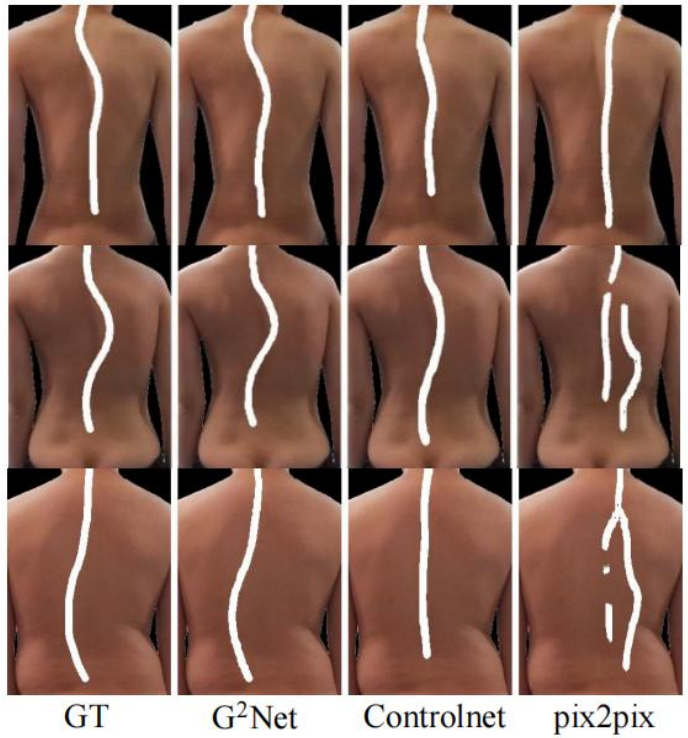


Detailed framework of the SpineLoRANet. The LoRABlock approximates weight updates by introducing two low-rank matrices, LoRA-A and LoRA-B. LoRA-A projects the input into a low-dimensional space, whilst LoRA-B projects it back to the original dimension, thereby enabling parameter-efficient adaptation to downstream tasks

Research project



Visual comparison of back-surface depth estimation results.



Visual comparison of spinal morphology estimation using different models.

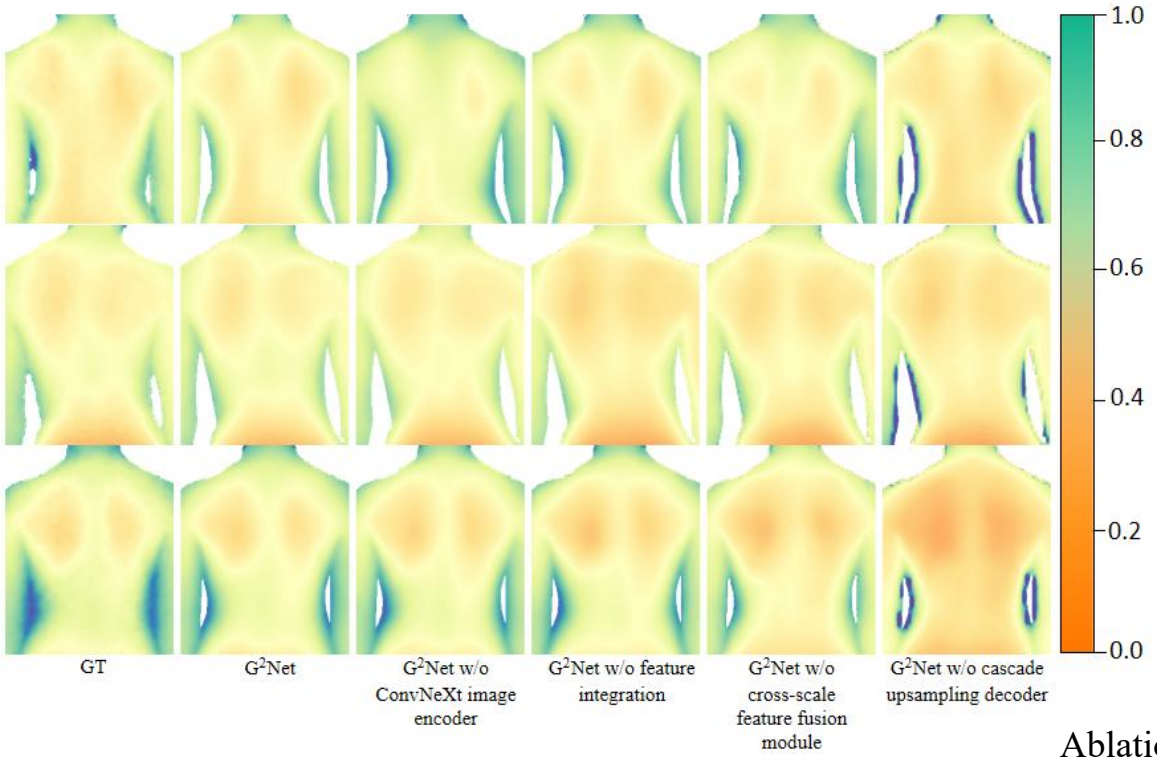
TABLE I: Quantitative Comparison of Monocular Depth Estimation Models

Model	$\epsilon_{\mathrm{ARE}}\downarrow$	$\epsilon_{\mathrm{RMSE}}\downarrow$	$\epsilon_{\mathrm{SRE}}\downarrow$	$\delta_1\uparrow$	$\delta_2\uparrow$	$\delta_3\uparrow$
FDE [56]	0.204	0.129	0.046	0.658	0.877	0.960
FDSI [57]	0.142	0.091	0.028	0.805	0.944	0.985
Marigold-DC [58]	0.193	0.088	0.084	0.930	0.977	0.981
G²Net	**0.053**	**0.048**	**0.006**	**0.962**	**0.990**	**0.993**

TABLE II: Performance Comparison of Spinal Morphology Estimation Models

Model	SSIM↑		Length diff.↓		Curvature diff.↓	
	mean	SD	mean	SD	mean	SD
ControlNet [48]	0.949	0.012	0.131	0.212	0.051	0.175
pix2pix [59]	0.914	0.015	0.196	0.167	0.030	0.021
G²Net	**0.950**	**0.013**	**0.079**	**0.148**	**0.028**	**0.112**

Research project



- Removing the image encoder would severely damage the core deep representation, resulting in a decline in prediction accuracy, and causing significant deviations at anatomical boundaries.
- Removing the Mamba-based feature integrator would cause discontinuity in depth transition, and blur the representation in local areas (especially the lumbar region), highlighting the crucial role of this module in cross-level feature fusion.
- Omitting the cross-scale feature fusion module would disrupt the depth consistency between multiple scales, verifying its core value in enhancing semantic-geometric synergy.
- The variant without the cascade upsampling decoder shows the most obvious artifacts and depth prediction misalignment in the back and lumbar regions, revealing the importance of the decoder in progressive resolution restoration and contour fidelity.

Ablation Study on the Components of our model

Model Variant	Components				Metrics					
	ConvNeXt image encoder	Mamba-based feature integrator	Cross-scale feature fusion module	Cascade upsampling decoder	ε_{ARE} ↓	$\varepsilon_{\mathrm{RMSE}}$ ↓	$\varepsilon_{\mathrm{SRE}}$ ↓	δ1 ↑	δ2 ↑	δ3 ↑
w/o ConvNeXt image encoder	✗	✓	✓	✓	0.690	0.251	0.332	0.538	0.702	0.788
w/o Mamba-based feature integrator	✓	✗	✓	✓	0.714	0.255	0.342	0.531	0.695	0.782
w/o Cross-scale feature fusion module	✓	✓	✗	✓	0.698	0.256	0.334	0.524	0.690	0.778
w/o Cascade upsampling decoder	✓	✓	✓	✗	0.707	0.182	0.166	0.558	0.730	0.796
G²Net (full model)	✓	✓	✓	✓	0.158	0.060	0.011	0.821	0.940	0.976

Research project

Thesis Publication

Journal: Chaos, Solitons and Fractals JCR Rank: SCI Q1 Top Journal

2026 JCR Impact Factor: 5.7

Title: MDHGFN: Multiscale Dual Hypergraph Fusion Spatiotemporal Network for Traffic Flow Prediction

Invention Patent

Patent Title:一种基于多尺度对偶超图融合的交通流预测方法

Application Number:202510759638.1



Chaos, Solitons and Fractals

Contents lists available

Chaos, Solitons a

journal homepage: www.elsev

MDHGFN: Multiscale Dual Hypergraph Fus
Network for traffic flow prediction

Jiajun Xian, Yixuan Ye, Wei Zhang, Zishen Che
Zichun Lin, Can Zhou, Hao LIU, Dan Yang, *,
Teng Zhou

ARTICLE INFO

ABSTRACT

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发文日：

2026年05月19日

申请号：202510759638.1 发文序号：2026051900238210

申请人：电子科技大学长三角研究院（湖州）

发明创造名称：一种基于多尺度对偶超图融合的交通流预测方法

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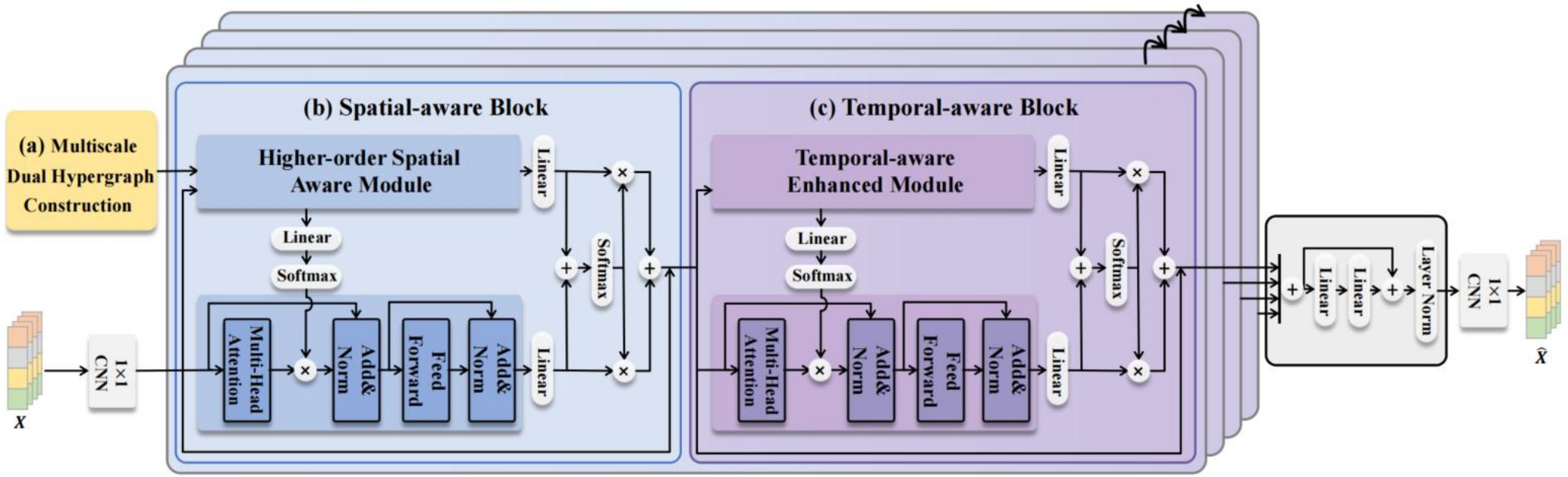
https://doi.org/10.1016/j.chaos.2025.117228
Received 1 June 2025; Received in revised form 27 July 2025; Accepted 11 September 2025
Available online 24 September 2025
0960-0779/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Research project

➢ **Model architecture diagram**

Transportation Topology Map
G = (V, E, A)

Traffic Feature Matrix
$X \in R_{T \times N \times D}$



Historical traffic flow data → Multiscale Dual Hypergraph Construction Module → Spatial perception module → Time perception module → Traffic flow prediction results

Research project

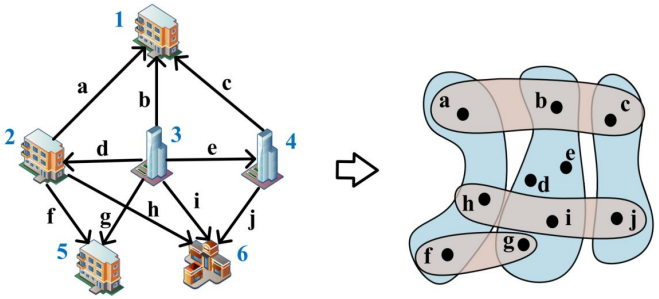
➢ Microscopic scale - Travel correlation dual hypergraph $G_t = (V_t, \varepsilon^t, H_t)$

✓ Construct the origin hyper-edge connection relationship for all nodes that share the same road node as the starting point.H_{src}^t

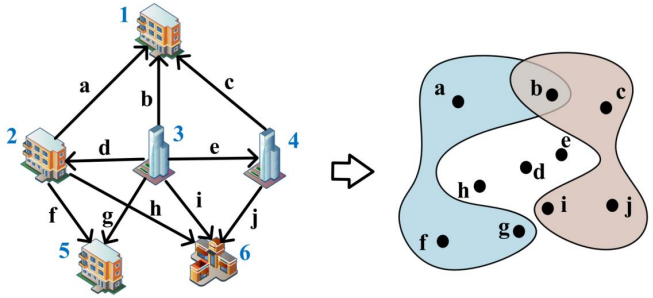
✓ Construct destination hyper-edge connection relationships for all nodes that share the same road node as the destination.H_{dst}^t

✓ H_{src}^t and H_{dst}^t together constitute the complete topological structure of the travel-related dual hypergraph.H_t

(a) Travel Intent Dual Hypergraph



(b) Functional Community Dual Hypergraph



➢ Mesoscopic scale - Functional community dual hypergraph $G_c = (V_c, \varepsilon^c, H_c)$

✓ Based on the original adjacency matrix, the functional similarity $A_{i,j}^c$ between nodes (with smaller values indicating greater similarity) is calculated.

✓ The K-nearest neighbor algorithm is used to select the c most functionally similar neighbors for each node (origin and destination) to form functional community hyperedges $e_i^c \in \varepsilon^c$. At the same time, the spatial distance between nodes within the hyperedge is constrained to be the shortest.

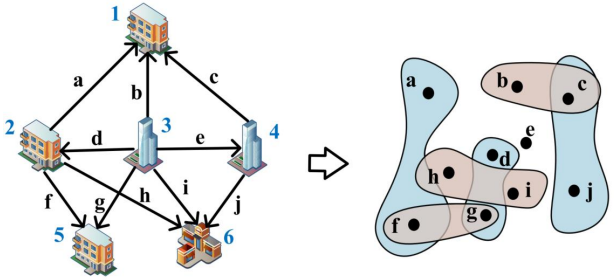
✓ H_{src}^c and H_{dst}^c together constitute the complete topological structure of the functional community dual hypergraph, H_c

➢ Macroscopic scale - Regional conduction dual hypergraph $G_r = (V_r, \varepsilon^r, H_r)$

✓ Through Top-K sampling, select the top K most influential starting and ending neighbors for each node, construct the node influence matrix Ar, and combine the Top-K influence neighbor sets of each node as hyperedges $e_i^r \in \varepsilon^r$

✓ H_{src}^r and H_{dst}^r together constitute the complete topological structure of the regional conduction dual hypergraph, H_r

(c) Regional Propagation Dual Hypergraph



Research project

- **Dataset:** PEMS03, PEMS04, PEMS07 and PEMS08, collected by the California Department of Transportation through its Highway Performance Measurement System (PEMS), and have been widely applied in the field of traffic flow research.

Statistical information of the dataset

Dataset	Nodes Number	Edges Number	Time Steps	Date Range
PEMS03	358	547	26208	09/01/2018 - 11/30/2018
PEMS04	307	340	16992	01/01/2018 - 02/28/2018
PEMS07	883	866	28224	05/01/2017 - 08/31/2017
PEMS08	170	295	17856	07/01/2016 - 08/31/2016

The dataset was divided into 60% for training, 20% for validation, and 20% for testing, which is consistent with the baseline method.

Evaluation metrics: Three widely used evaluation metrics were adopted to assess the performance of the model: Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Root Mean Square Error (RMSE).

- **Baseline:** 14 baseline methods were selected for comparison experiments with the proposed method, and the parameter settings recommended in the original paper were followed to ensure the reliability and fairness of the comparison results.

Baseline Model Comparison Summary

Model Abbreviation	Publication Year	Core Innovation
FC-LSTM	2015	Fully connected LSTM structure focusing on time-series modeling
DCRNN	2017	Bidirectional random walk with an encoder-decoder framework
STGCN	2018	Fully convolutional graph structure for efficient spatio-temporal modeling
Graph WaveNet	2019	Adaptive dependency matrix and dilated convolution
STSGCN	2020	Synchronous spatio-temporal modeling and temporal heterogeneity handling
OGCRNN	2020	Data-driven graph structure optimization
STGODE	2021	Tensor differential equation and semantic adjacency matrix
STFGNN	2021	Spatio-temporal graph fusion and gated convolution
GMSDR	2022	Explicit modeling of multi-step dependencies
FEDformer	2022	Frequency-domain enhancement and seasonal-trend decomposition
HSTGCNT	2023	Integration of hierarchical Transformer and hypergraph convolution
AGFCRN	2024	Adaptive hypergraph fusion and attention mechanism
DS-STGCN	2024	Dynamic multi-scale spatio-temporal modeling
CDAGF	2024	Contrastive learning and residual-enhanced decomposition

Research project

① Traditional sequential modeling methods failed to effectively model spatial dependence.

② The introduction of graph structures to capture spatial characteristics still has shortcomings in modeling long-term sequence dependence and dynamic spatial relationships.

③ Its performance on large-scale datasets remains limited, and it is difficult to comprehensively capture the complex spatiotemporal characteristics.

④ It performs well in long-term sequence prediction, but still has certain limitations in modeling complex spatial dependencies.

Performance comparison results of different baseline models and MDGFN on each dataset

Method		PEMS03			PEMS04			PEMS07			PEMS08			Average(Rank)		
		MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)
2015	FC-LSTM [50]	21.33	35.11	23.33	27.14	18.20	18.20	29.98	45.94	13.20	22.20	34.06	14.20	12	7	11
2017	DCRNN [51]	24.11	38.69	22.67	30.49	45.99	20.72	34.82	51.32	16.23	21.76	32.41	14.19	14	14	13
2018	STGCN [40]	23.48	37.51	21.95	28.80	44.00	19.08	33.80	51.74	14.75	22.49	34.05	14.54	13	13	12
2019	Graph WaveNet [52]	24.73	38.42	24.01	30.55	46.31	23.33	34.62	51.86	15.68	23.01	35.67	15.37	15	15	14
2020	STSGCN [53]	20.72	33.75	19.44	24.21	38.36	15.63	28.10	45.88	11.90	19.56	30.58	12.55	10	12	9
2020	OGCRNN [54]	17.16	29.40	16.48	22.16	35.15	15.72	24.45	39.80	10.22	18.50	29.00	11.74	7	8	7
2021	STGODE [55]	16.50	27.84	16.69	20.84	32.82	13.77	22.59	37.54	10.14	16.81	25.97	10.62	4	4	5
2021	STFGNN [56]	19.37	32.57	18.37	21.55	34.60	14.26	24.77	40.55	10.33	18.83	29.66	11.90	8	10	8
2022	GMSDR [57]	15.92	28.87	14.50	21.63	34.05	13.87	23.78	44.60	10.17	16.59	25.69	10.59	6	6	4
2022	FEDformer [58]	19.74	29.51	20.04	23.57	35.65	50.54	29.42	43.65	22.85	24.22	35.32	17.38	11	11	15
2023	HSTGCNT [59]	19.22	30.65	19.72	23.47	35.53	16.52	24.43	37.77	11.25	20.82	30.87	13.76	9	9	10
2023	AGFCRN [60]	16.89	28.98	17.65	20.51	33.81	13.81	22.69	37.95	9.62	17.35	27.66	11.31	5	5	6
2024	DS-STGCN [14]	16.04	26.40	15.69	19.80	31.81	13.40	22.11	36.62	9.38	15.66	25.65	**10.09**	3	2	2
2024	CDAGF [61]	16.54	27.71	16.28	19.55	32.58	13.06	21.56	37.03	9.11	15.68	25.71	10.13	2	3	3
(Ours)	**MDHGFN**	**14.12**	**22.52**	**12.95**	**19.29**	**31.57**	**12.35**	**20.41**	**33.69**	**8.98**	**15.45**	**25.20**	11.50	1	1	1

⑤ Due to the limitations of the low-order graph structure and the expressive capacity of the single-scale hypergraph, it is impossible to fully extract the multi-scale high-order spatial information contained in the traffic flow data.

The evaluation results on all datasets have reached the current optimal level, and the performance on large-scale datasets is superior to that on small-scale datasets.

Research project

➢ Module ablation experiment

The ablation experiment results of different model variants on the PEMS03 and PEMS08 datasets

Datasets	Model	RMSE	MAE	MAPE (%)
PEMS03	MDHGFN w/o both HSAM and TAEM	23.48	14.91	13.83
PEMS03	MDHGFN w/o HSAM	23.07	14.64	13.51
PEMS03	MDHGFN w/o TAEM	22.68	14.21	13.39
PEMS03	MDHGFN	22.52	14.12	12.95
PEMS08	MDHGFN w/o both HSAM and TAEM	26.54	16.72	12.93
PEMS08	MDHGFN w/o HSAM	25.57	15.90	11.89
PEMS08	MDHGFN w/o TAEM	25.26	15.52	11.82
PEMS08	MDHGFN	25.20	15.45	11.50

The experimental results show that the complete model outperforms all the ablation variants in terms of RMS E, MAE and MAPE metrics, thereby verifying the crucial role of the HSAM and TAEM modules in the modeling of spatio-temporal characteristics.

➢ Supergraph ablation experiment

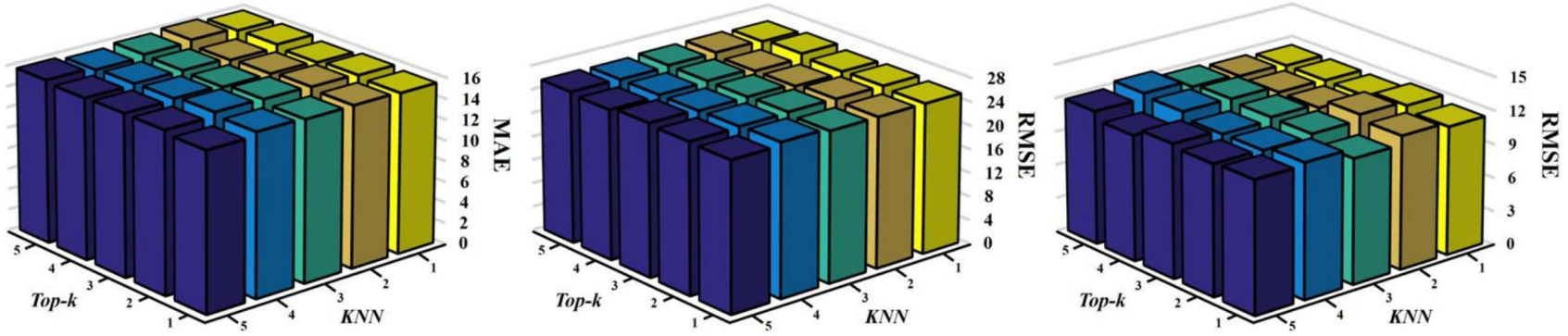
Joint learning of multi-scale dual hypergraphs can effectively integrate high-order spatial information at different scales, significantly improving the performance of the model.

The ablation experiment results of different-scale dual hypergraph combinations on the PEMS03 and PEMS08 datasets

Datasets	Dual Hypergraph	RMSE	MAE	MAPE (%)
PEMS03	$[G^r]$	22.63	14.23	13.01
PEMS03	$[G^c]$	22.66	14.30	13.48
PEMS03	$[G^t]$	22.88	14.53	15.03
PEMS03	$[G^r, G^c, G^t]$	22.52	14.12	12.95
PEMS08	$[G^r]$	25.49	15.48	11.52
PEMS08	$[G^c]$	25.36	15.53	12.07
PEMS08	$[G^t]$	25.52	15.46	11.57
PEMS08	$[G^r, G^c, G^t]$	25.20	15.45	11.50

Research project

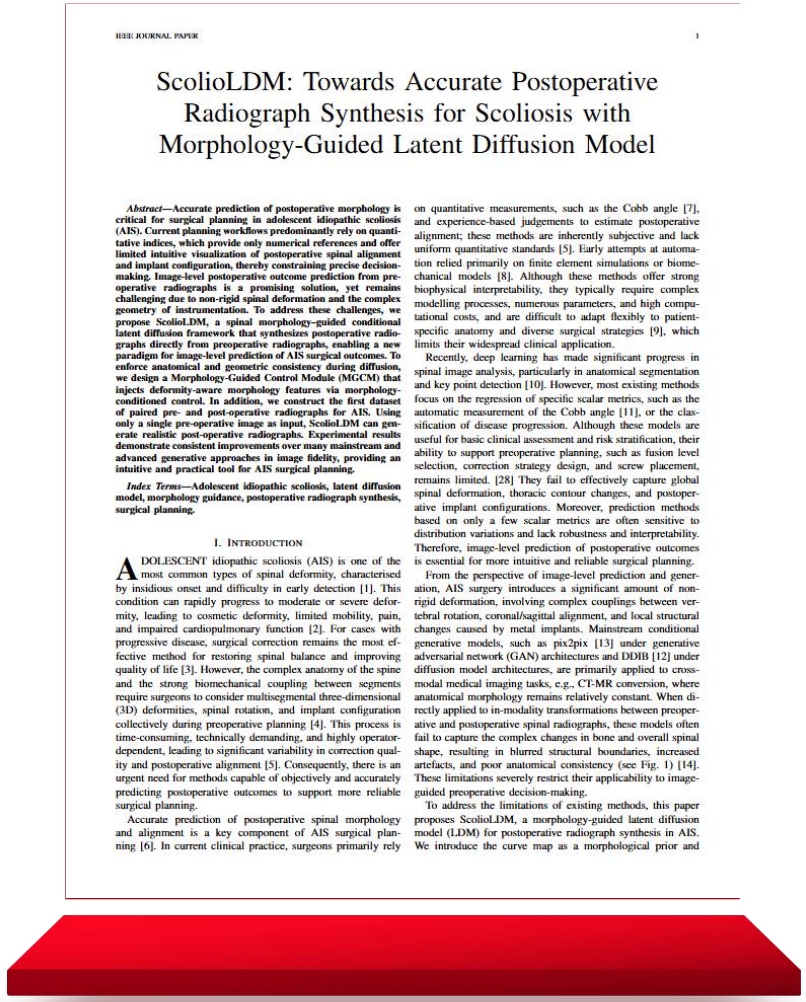
➢ **Hyperparameter experiment result graph**



Evaluate the impact of different K values (the number of KNN neighbors used for the functional community dual hypergraph) and Top-k values (the number of key nodes selected for the regional conduction dual hypergraph) on the model performance

- ✓ A smaller K value may lead to insufficient community division, failing to capture the extensive interaction information between regions; while a larger K value introduces too many irrelevant nodes, resulting in redundancy and affecting the extraction of spatial features.
- ✓ Similarly, a smaller Top-k value may ignore the critical path dependency information, while an overly large Top-k value will introduce irrelevant paths and interfere with the modeling of critical dependency relationships.
- ✓ Based on the comprehensive experimental results, appropriately setting the values of K and Top-k can effectively balance the feature sufficiency and redundancy, fully leveraging the advantages of the functional community dual hypergraph and the regional conduction dual hypergraph, thereby significantly enhancing the model's ability to model high-order spatiotemporal relationships.

Research project



IEEE JOURNAL PAPER 1

ScolioLDM: Towards Accurate Postoperative Radiograph Synthesis for Scoliosis with Morphology-Guided Latent Diffusion Model

***Abstract*—Accurate prediction of postoperative morphology is critical for surgical planning in adolescent idiopathic scoliosis (AIS). Current planning workflows predominantly rely on quantitative indices, which provide only numerical references and offer limited intuitive visualization of postoperative spinal alignment and implant configuration, thereby constraining precise decision-making. Image-level postoperative outcome prediction from preoperative radiographs is a promising solution, yet remains challenging due to non-rigid spinal deformation and the complex geometry of instrumentation. To address these challenges, we propose ScolioLDM, a spinal morphology–guided conditional latent diffusion framework that synthesizes postoperative radiographs directly from preoperative radiographs, enabling a new paradigm for image-level prediction of AIS surgical outcomes. To enforce anatomical and geometric consistency during diffusion, we design a Morphology-Guided Control Module (MGCM) that injects deformity-aware morphology features via morphology-conditioned control. In addition, we construct the first dataset of paired pre- and post-operative radiographs for AIS. Using only a single pre-operative image as input, ScolioLDM can generate realistic post-operative radiographs. Experimental results demonstrate consistent improvements over many mainstream and advanced generative approaches in image fidelity, providing an intuitive and practical tool for AIS surgical planning.**

***Index Terms*—Adolescent idiopathic scoliosis, latent diffusion model, morphology guidance, postoperative radiograph synthesis, surgical planning.**

I. Introduction

ADOLESCENT idiopathic scoliosis (AIS) is one of the most common types of spinal deformity, characterised by insidious onset and difficulty in early detection [1]. This condition can rapidly progress to moderate or severe deformity, leading to cosmetic deformity, limited mobility, pain, and impaired cardiopulmonary function [2]. For cases with progressive disease, surgical correction remains the most effective method for restoring spinal balance and improving quality of life [3]. However, the complex anatomy of the spine and the strong biomechanical coupling between segments require surgeons to consider multisegmental three-dimensional (3D) deformities, spinal rotation, and implant configuration collectively during preoperative planning [4]. This process is time-consuming, technically demanding, and highly operator-dependent, leading to significant variability in correction quality and postoperative alignment [5]. Consequently, there is an urgent need for methods capable of objectively and accurately predicting postoperative outcomes to support more reliable surgical planning.

Accurate prediction of postoperative spinal morphology and alignment is a key component of AIS surgical planning [6]. In current clinical practice, surgeons primarily rely on quantitative measurements, such as the Cobb angle [7], and experience-based judgements to estimate postoperative alignment; these methods are inherently subjective and lack uniform quantitative standards [5]. Early attempts at automation relied primarily on finite element simulations or biomechanical models [8]. Although these methods offer strong biophysical interpretability, they typically require complex modelling processes, numerous parameters, and high computational costs, and are difficult to adapt flexibly to patient-specific anatomy and diverse surgical strategies [9], which limits their widespread clinical application.

Recently, deep learning has made significant progress in spinal image analysis, particularly in anatomical segmentation and key point detection [10]. However, most existing methods focus on the regression of specific scalar metrics, such as the automatic measurement of the Cobb angle [11], or the classification of disease progression. Although these models are useful for basic clinical assessment and risk stratification, their ability to support preoperative planning, such as fusion level selection, correction strategy design, and screw placement, remains limited. [28] They fail to effectively capture global spinal deformation, thoracic contour changes, and postoperative implant configurations. Moreover, prediction methods based on only a few scalar metrics are often sensitive to distribution variations and lack robustness and interpretability. Therefore, image-level prediction of postoperative outcomes is essential for more intuitive and reliable surgical planning.

From the perspective of image-level prediction and generation, AIS surgery introduces a significant amount of non-rigid deformation, involving complex couplings between vertebral rotation, coronal/sagittal alignment, and local structural changes caused by metal implants. Mainstream conditional generative models, such as pix2pix [13] under generative adversarial network (GAN) architectures and DDIB [12] under diffusion model architectures, are primarily applied to cross-modal medical imaging tasks, e.g., CT-MR conversion, where anatomical morphology remains relatively constant. When directly applied to in-modality transformations between preoperative and postoperative spinal radiographs, these models often fail to capture the complex changes in bone and overall spinal shape, resulting in blurred structural boundaries, increased artefacts, and poor anatomical consistency (see Fig. 1) [14]. These limitations severely restrict their applicability to image-guided preoperative decision-making.

To address the limitations of existing methods, this paper proposes ScolioLDM, a morphology-guided latent diffusion model (LDM) for postoperative radiograph synthesis in AIS. We introduce the curve map as a morphological prior and

Title of the paper

ScolioLDM: Towards Accurate Postoperative Radiograph Synthesis for Scoliosis with Morphology-Guided Latent Diffusion Model

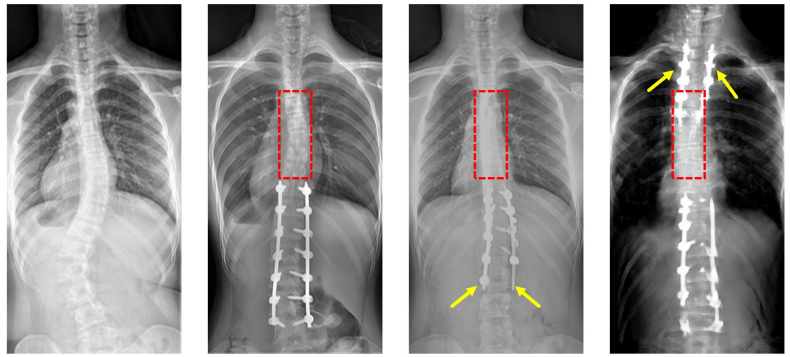
Research content

This work designs ScolioLDM, which uses predicted spine contour maps and a MGCM to guide latent diffusion for high-fidelity AIS postoperative X-ray synthesis, validated on a 612-patient clinical dataset.

Research results

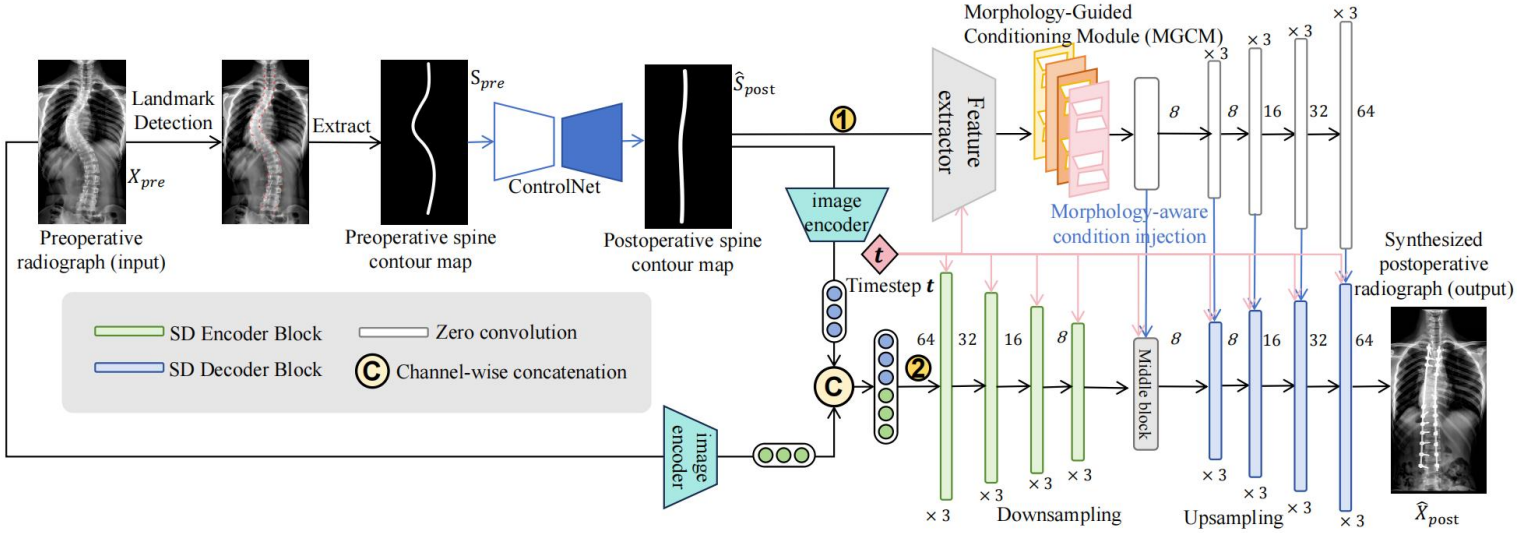
Experiments on 612 AIS cases prove ScolioLDM outperforms mainstream generative models, achieving top PSNR and NCC. Ablations verify spine contour maps and MGCM are vital for realistic spine and implant reconstruction.

Research project

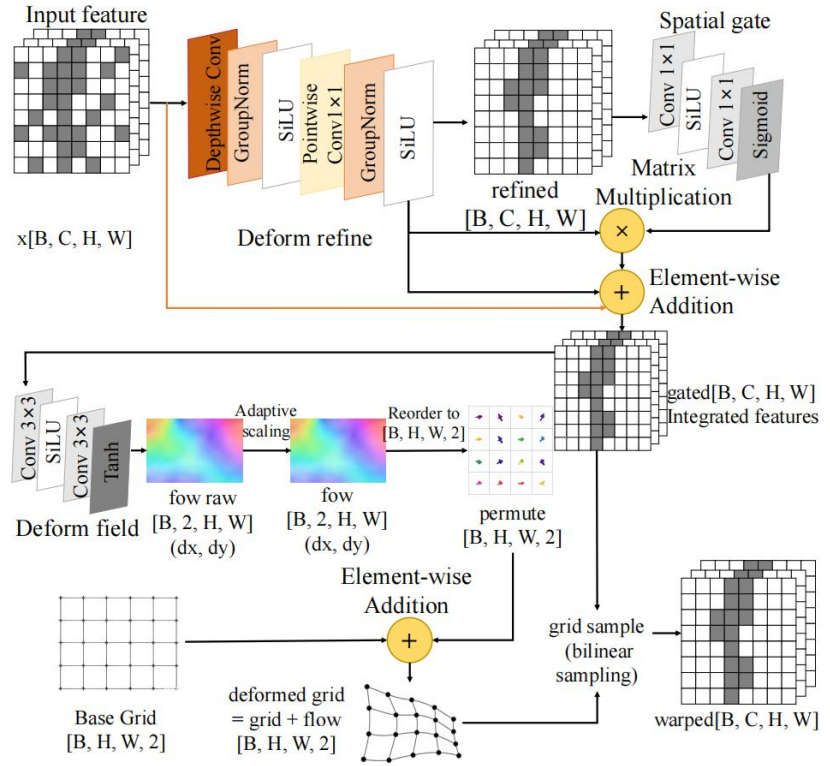


(a) preoperative radiograph (b) postoperative radiograph (GT) (c) DDIB (d) pix2pix

Illustration of postoperative radiograph synthesis directly using mainstream conditional generative models.When directly applied to in-modality transformations between preoperative and postoperative spinal radiographs, these models often fail to capture the complex changes in bone and overall spinal shape, resulting in blurred structural boundaries, increased artefacts, and poor anatomical consistency.

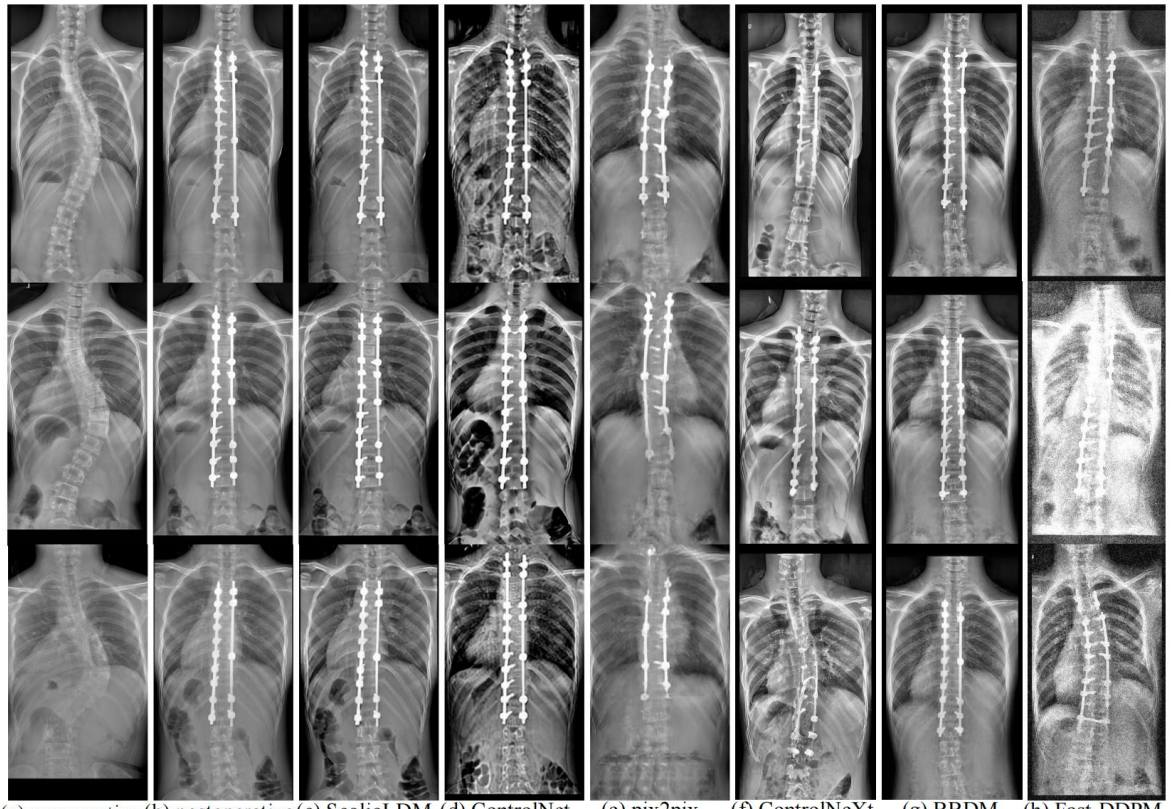


Overview of the proposed ScolioLDM. The left part presents the overall framework for postoperative radiograph synthesis, while the right part illustrates the detailed architecture of the core denoising U-Net integrated with the MGCM. ControlNet is used to introduce additional structural conditions into the diffusion model for controllable postoperative radiograph synthesis .



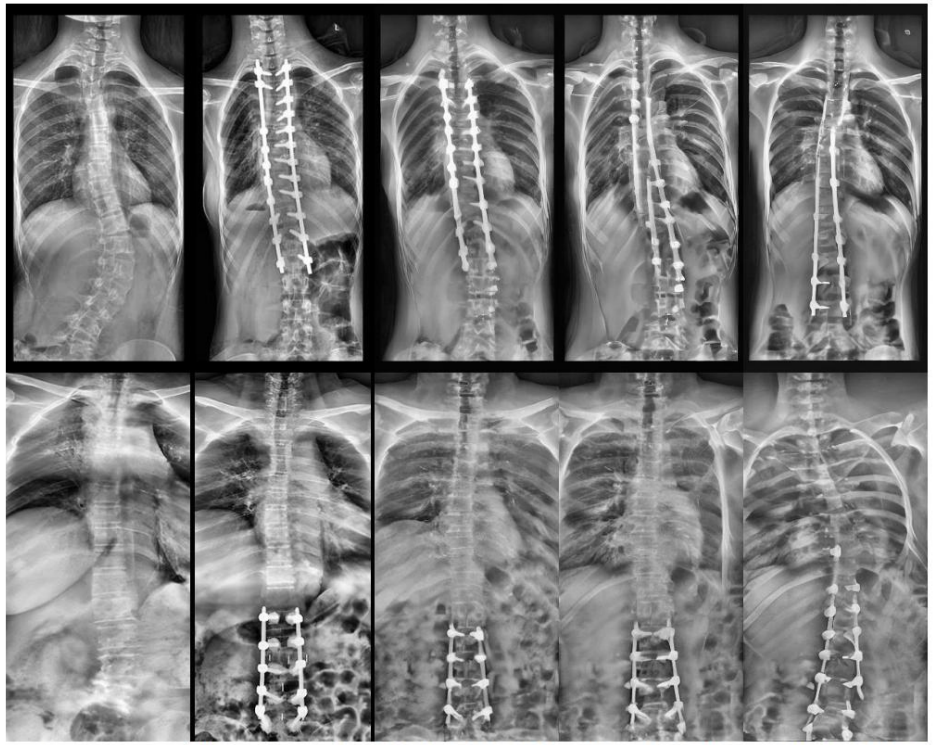
Detailed illustration of the Morphology-Guided Control Module (MGCM) embedded in the proposed ScolioLDM framework.

Research project



(a) preoperative radiograph (b) postoperative radiograph (GT) (c) ScolioLDM (d) ControlNet (e) pix2pix (f) ControlNeXt (g) BBDM (h) Fast-DDPM

Overall, both the quantitative and qualitative results demonstrate that the proposed framework can synthesize postoperative radiographs with high anatomical fidelity, structural consistency, and visual realism, while also confirming that MGCM and postoperative curve map guidance are essential components that significantly enhance the anatomical accuracy and visual quality of the synthesized results.

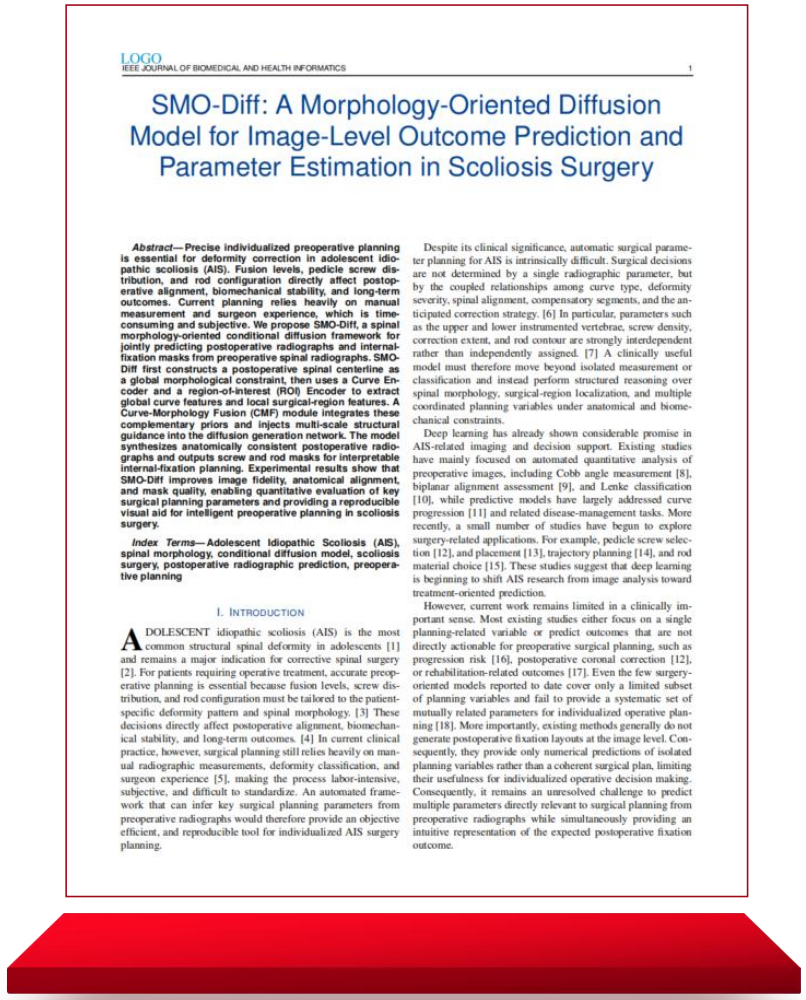


(a) preoperative radiograph (b) postoperative radiograph (GT) (c) ScolioLDM (d) ScolioLDM w/o MGCM (e) ScolioLDM w/o curve map

Method	SSIM ↑	PSNR ↑	MAE ↓	GradMAE ↓	NCC ↑
ControlNet	0.831 ± 0.009	19.726 ± 1.121	0.036 ± 0.005	0.054 ± 0.005	0.877 ± 0.03
pix2pix	0.846 ± 0.007	19.432 ± 0.861	0.035 ± 0.004	0.061 ± 0.006	0.861 ± 0.03
ControlNeXt	0.841 ± 0.010	19.179 ± 1.140	0.037 ± 0.006	0.059 ± 0.007	0.864 ± 0.03
BBDM	0.856 ± 0.010	19.654 ± 1.118	0.034 ± 0.005	0.053 ± 0.006	0.869 ± 0.03
Fast-DDPM	0.810 ± 0.011	17.539 ± 1.987	0.049 ± 0.015	0.067 ± 0.012	0.843 ± 0.06
ScolioLDM	0.841 ± 0.011	20.436 ± 1.322	0.032 ± 0.006	0.051 ± 0.006	0.892 ± 0.03

Variant	SSIM ↑	PSNR ↑	MAE ↓	GradMAE ↓	NCC ↑
w/o MGCM	0.836 ± 0.009	19.951 ± 1.142	0.034 ± 0.005	0.053 ± 0.005	0.882 ± 0.035
w/o postoperative curve map	0.822 ± 0.008	19.226 ± 1.007	0.039 ± 0.005	0.057 ± 0.006	0.857 ± 0.039
ScolioLDM	0.841 ± 0.011	20.436 ± 1.322	0.032 ± 0.006	0.051 ± 0.006	0.892 ± 0.035

Research project



LOGO
IEEE JOURNAL OF BIOMEDICAL AND HEALTH INFORMATICS 1

SMO-Diff: A Morphology-Oriented Diffusion Model for Image-Level Outcome Prediction and Parameter Estimation in Scoliosis Surgery

***Abstract*— Precise individualized preoperative planning is essential for deformity correction in adolescent idiopathic scoliosis (AIS). Fusion levels, pedicle screw distribution, and rod configuration directly affect postoperative alignment, biomechanical stability, and long-term outcomes. Current planning relies heavily on manual measurement and surgeon experience, which is time-consuming and subjective. We propose SMO-Diff, a spinal morphology-oriented conditional diffusion framework for jointly predicting postoperative radiographs and internal-fixation masks from preoperative spinal radiographs. SMO-Diff first constructs a postoperative spinal centerline as a global morphological constraint, then uses a Curve Encoder and a region-of-interest (ROI) Encoder to extract global curve features and local surgical-region features. A Curve-Morphology Fusion (CMF) module integrates these complementary priors and injects multi-scale structural guidance into the diffusion generation network. The model synthesizes anatomically consistent postoperative radiographs and outputs screw and rod masks for interpretable internal-fixation planning. Experimental results show that SMO-Diff improves image fidelity, anatomical alignment, and mask quality, enabling quantitative evaluation of key surgical planning parameters and providing a reproducible visual aid for intelligent preoperative planning in scoliosis surgery.**

***Index Terms*— Adolescent Idiopathic Scoliosis (AIS), spinal morphology, conditional diffusion model, scoliosis surgery, postoperative radiographic prediction, preoperative planning**

I. Introduction

ADOLESCENT idiopathic scoliosis (AIS) is the most common structural spinal deformity in adolescents [1] and remains a major indication for corrective spinal surgery [2]. For patients requiring operative treatment, accurate preoperative planning is essential because fusion levels, screw distribution, and rod configuration must be tailored to the patient-specific deformity pattern and spinal morphology. [3] These decisions directly affect postoperative alignment, biomechanical stability, and long-term outcomes. [4] In current clinical practice, however, surgical planning still relies heavily on manual radiographic measurements, deformity classification, and surgeon experience [5], making the process labor-intensive, subjective, and difficult to standardize. An automated framework that can infer key surgical planning parameters from preoperative radiographs would therefore provide an objective efficient, and reproducible tool for individualized AIS surgery planning.

Despite its clinical significance, automatic surgical parameter planning for AIS is intrinsically difficult. Surgical decisions are not determined by a single radiographic parameter, but by the coupled relationships among curve type, deformity severity, spinal alignment, compensatory segments, and the anticipated correction strategy. [6] In particular, parameters such as the upper and lower instrumented vertebrae, screw density, correction extent, and rod contour are strongly interdependent rather than independently assigned. [7] A clinically useful model must therefore move beyond isolated measurement or classification and instead perform structured reasoning over spinal morphology, surgical-region localization, and multiple coordinated planning variables under anatomical and biomechanical constraints.

Deep learning has already shown considerable promise in AIS-related imaging and decision support. Existing studies have mainly focused on automated quantitative analysis of preoperative images, including Cobb angle measurement [8], biplanar alignment assessment [9], and Lenke classification [10], while predictive models have largely addressed curve progression [11] and related disease-management tasks. More recently, a small number of studies have begun to explore surgery-related applications. For example, pedicle screw selection [12], and placement [13], trajectory planning [14], and rod material choice [15]. These studies suggest that deep learning is beginning to shift AIS research from image analysis toward treatment-oriented prediction.

However, current work remains limited in a clinically important sense. Most existing studies either focus on a single planning-related variable or predict outcomes that are not directly actionable for preoperative surgical planning, such as progression risk [16], postoperative coronal correction [12], or rehabilitation-related outcomes [17]. Even the few surgery-oriented models reported to date cover only a limited subset of planning variables and fail to provide a systematic set of mutually related parameters for individualized operative planning [18]. More importantly, existing methods generally do not generate postoperative fixation layouts at the image level. Consequently, they provide only numerical predictions of isolated planning variables rather than a coherent surgical plan, limiting their usefulness for individualized operative decision making. Consequently, it remains an unresolved challenge to predict multiple parameters directly relevant to surgical planning from preoperative radiographs while simultaneously providing an intuitive representation of the expected postoperative fixation outcome.

Title of the paper

SMO-Diff: A Morphology-Oriented Diffusion Model for Image-Level Outcome Prediction and Parameter Estimation in Scoliosis Surgery

Research content

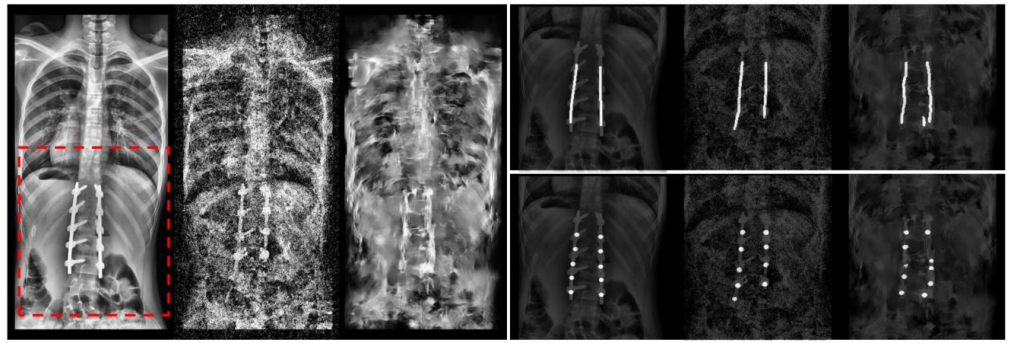
This paper proposes a spinal morphology-oriented conditional diffusion framework SMO-Diff to generate anatomically consistent postoperative spinal radiographs and visually interpretable segmentation masks of pedicle screws and connecting rods, supporting intuitive individualized preoperative planning for adolescent idiopathic scoliosis corrective surgery.

Research results

Quantitative metrics and visual comparisons on a clinical AIS dataset demonstrate that the proposed SMO-Diff outperforms mainstream generative models in postoperative radiograph synthesis and internal fixation mask reconstruction, with ablation experiments verifying the indispensable contributions of its Curve Encoder, ROI Encoder and CMF module.

Research project

Fig. 1. Representative performance of existing generative models for postoperative radiograph synthesis and internal-fixation reconstruction. (a) GT radiograph (red dashed box marks the surgical region). (b) Fast-DDPM produces severe noise. (c) pix2pix generates blurry anatomy. (d) GT rod and screw masks. (e) Fast-DDPM fails to reconstruct continuous rods. (f) pix2pix yields discontinuous rods and misaligned screws.



(a) postoperative radiograph (GT) (b) Fast-DDPM (c) pix2pix (d) rods and screws(GT) (e) rods and screws (Fast-DDPM) (f) rods and screws(pix2pix)

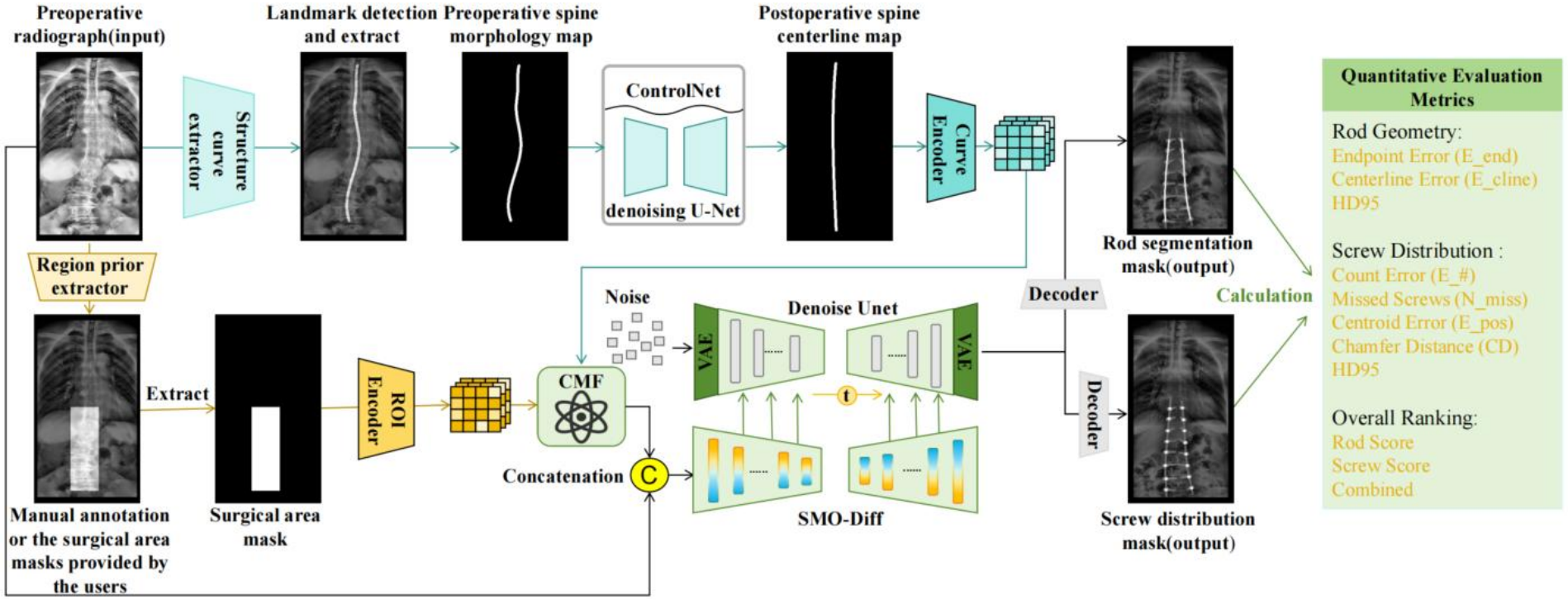


Fig. 2. Overview of the proposed SMO-Diff framework. The preoperative spinal centerline is first extracted and then fed into ControlNet to predict the postoperative centerline as a global morphological constraint. Together with the preoperative radiograph and the surgical region of interest, the predicted centerline guides the diffusion network to synthesize the postoperative radiograph and output the corresponding screw and rod masks. Specifically, the predicted postoperative centerline and the surgical region are encoded by the Curve Encoder and the ROI Encoder, respectively; their complementary priors are then integrated via a CMF module to inject multi-scale structural guidance into the denoising network (see Fig. 4 for CMF details).

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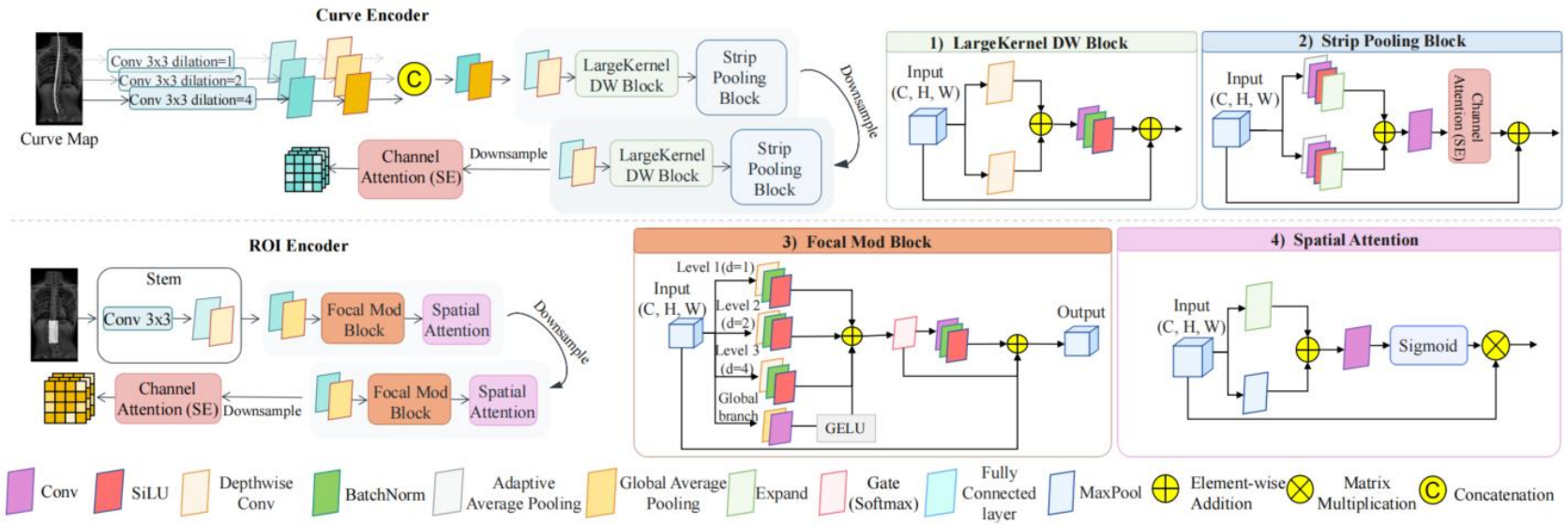


Fig. 3. Overview of the proposed Curve Encoder and ROI Encoder modules. The channel attention (SE) blocks implement squeeze-and-excitation mechanisms to adaptively recalibrate channel-wise feature responses, thereby emphasizing informative attributes and suppressing less relevant background responses.

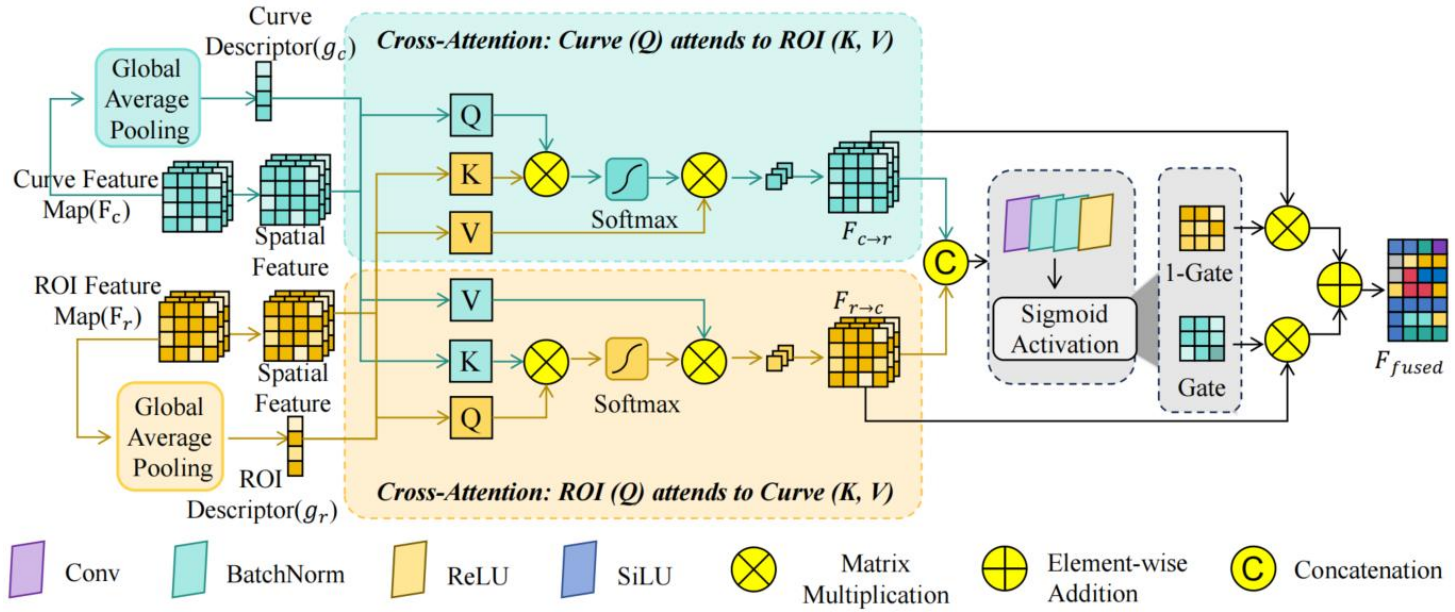


Fig. 4.Overview of the proposed CMF module

Research project

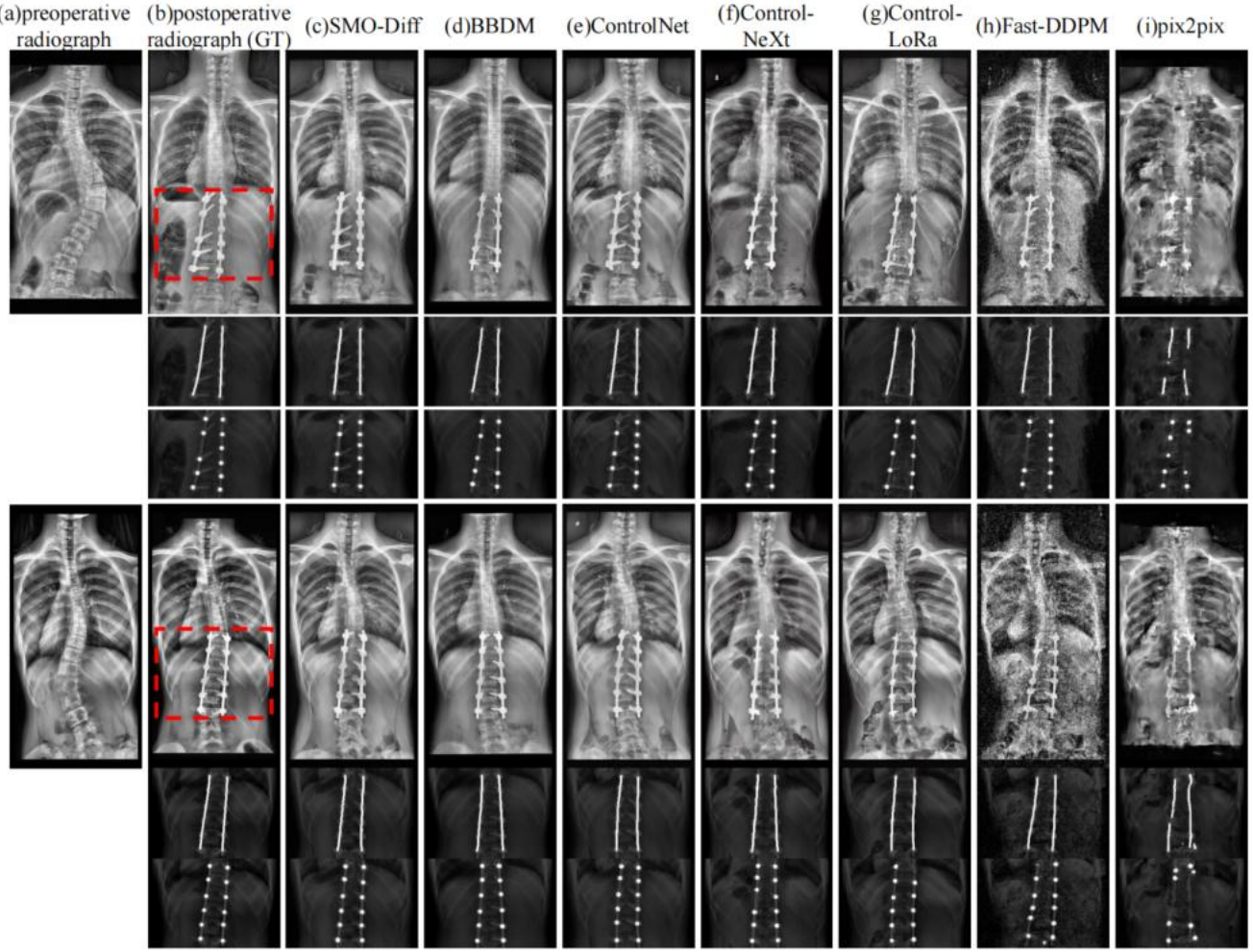


Fig. 5. Visual comparison of postoperative internal-fixation reconstruction across different methods on two representative cases. For each case, the rows show the reconstructed radiograph, rod placement, and pedicle screw distribution from top to bottom. Columns from left to right represent radiograph, postoperative radiograph (GT), SMO-Diff, BBDM, ControlNet, ControlNeXt, ControlLoRA, Fast-DDPM, pix2pix.

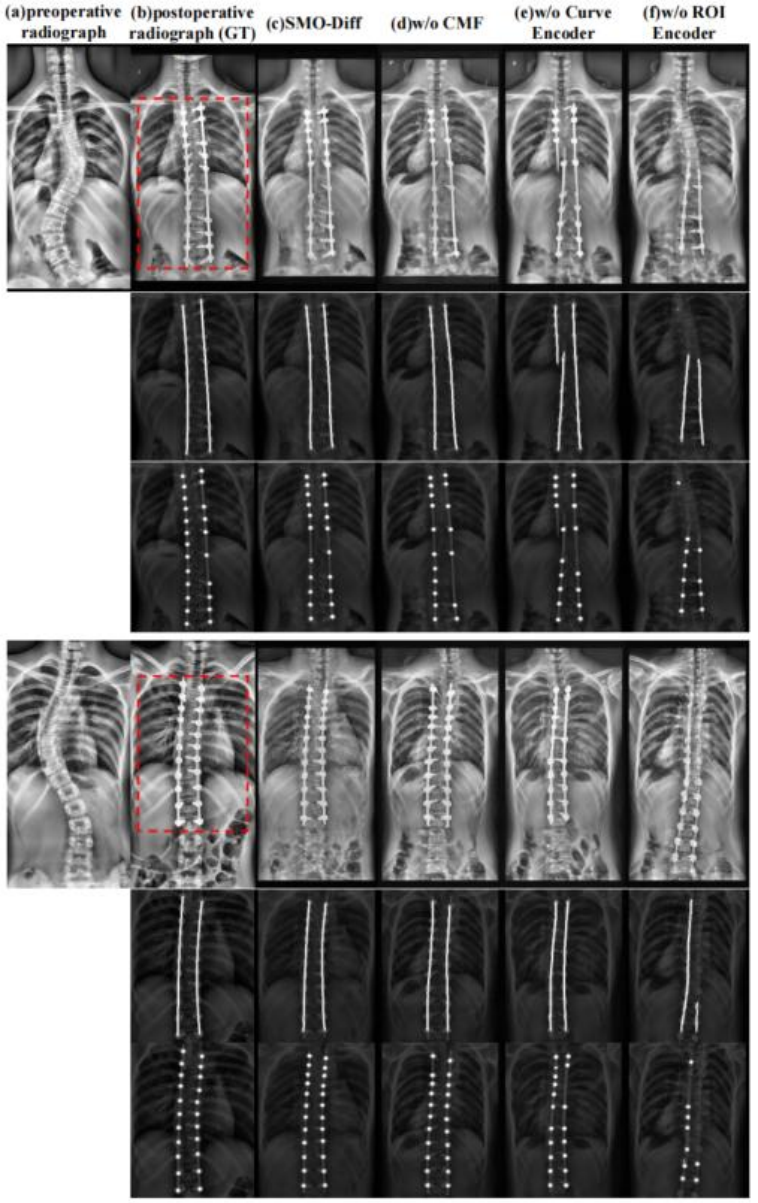


Fig. 6. Ablation study of key components in SMO-Diff on two representative cases. For each case, the rows show full postoperative reconstruction, rod placement, and pedicle screw distribution from top to bottom. The columns show: (a) preoperative radiograph, (b) postoperative radiograph (GT), (c) SMO-Diff, (d) SMO-Diff without CMF, (e) SMO-Diff without Curve Encoder, and (f) SMO-Diff without ROI Encoder.

Research project

TABLE I
COMPARISON OF QUANTITATIVE RESULTS ACROSS DIFFERENT ALGORITHM MODELS. ↓ (↑) INDICATES LOWER (HIGHER) IS BETTER.

Category	Metric	Symbol	SMO-Diff	Control-Net [21]	BBDM [42]	Control-NeXt [43]	Control-LoRA [44]	Fast-DDPM [20]	pix2pix [19]
Rod Geometry Measurement	Endpoint Error ↓	Eend (px)	**9.4**	10.1	10.3	11.8	15.9	12.5	51.5
	Centerline Error ↓	Ecline (px)	**3.9**	4.1	4.4	4.4	6.1	4.1	14.8
	HD95 (Centerline) ↓	HD95 (px)	**7.1**	8	8.1	9.7	15.5	12	57.4
Screw Distribution	Quantity Error ↓	E#	1.3	1.6	**1.1**	2.1	2	2	4
	Missing Screws ↓	Nmiss	**0.4**	0.9	0.6	1.1	1.2	1	3.8
	Centroid Error ↓	Epos (px)	**9.3**	10	10.4	11.3	11.1	10	12.6
	Chamfer Distance ↓	CD (px)	**9.2**	9.7	9.6	10.4	10.4	9.8	12.7
	HD95 (Centroid) ↓	HD95 (px)	**16.3**	17.7	18.5	20.5	20.2	19.6	29.6
Comprehensive Ranking	Rod Ranking Score ↓	Rrod	**1**	2	3.3	4.3	6	4.3	7
	Screw Ranking Score ↓	Rscrew	**1.2**	2.8	2.4	5.6	5.4	3.6	7
	Overall Ranking Score ↓	Roverall	**1.1**	2.4	2.9	5	5.7	4	7
Image Quality	SSIM ↑	–	0.61±0.03	0.59±0.02	**0.64±0.02**	0.63±0.02	0.57±0.02	0.53±0.02	0.61±0.10
	PSNR ↑	(dB)	**16.5±1.4**	15.9±1.2	15.9±1.1	16.0±1.1	15.2±1.2	13.5±1.8	15.4±0.9

TABLE II
QUANTITATIVE RESULTS OF MODULE ABLATION EXPERIMENTS FOR THE SMO-DIFF MODEL. ↓ (↑) INDICATES LOWER (HIGHER) IS BETTER.

Category	Metric	Symbol	SMO-Diff	w/o CMF	w/o Curve Encoder	w/o ROI Encoder
Rod Geometry Measurement	Endpoint Error ↓	Eend (px)	**9.4**	10.3	11.4	55.2
	Centerline Error ↓	Ecline (px)	**3.9**	4.4	4.9	19.7
	HD95 (Centerline) ↓	HD95 (px)	**7.1**	9.2	10.5	70.2
Screw Distribution	Quantity Error ↓	E#	**1.3**	1.9	1.9	6.2
	Missing Screws ↓	Nmiss	**0.4**	1.1	1.3	6.1
	Centroid Error ↓	Epos (px)	**9.3**	10.3	10.1	15
	Chamfer Distance ↓	CD (px)	**9.2**	10.1	10	18.6
	HD95 (Centroid) ↓	HD95 (px)	**16.3**	19	18.9	45.8
Comprehensive Ranking	Rod Ranking Score ↓	Rrod	**1**	2	3	4
	Screw Ranking Score ↓	Rscrew	**1**	2.7	2.3	4
	Overall Ranking Score ↓	Roverall	**1**	2.4	2.7	4
Image Quality	SSIM ↑	–	**0.61 ± 0.03**	0.57 ± 0.02	0.58 ± 0.02	0.57 ± 0.01
	PSNR ↑	dB	**16.50 ± 1.45**	15.91 ± 1.15	15.87 ± 1.21	15.55 ± 0.98

SMO-Diff outperforms all competitors across all quantitative metrics. For rod geometry, it achieves the lowest endpoint error (Eend=9.4 px), centerline error (Ecline=3.9 px), and HD95 (7.1 px), with the highest PSNR (16.50±1.45 dB) and SSIM (0.61±0.03). For screw distribution, it obtains the best Nmiss=0.4, Epos=9.3 px, CD= 9.2 px, and HD95= 16.3 px, ranking first in rod, screw, and overall scores. Though BBDM has the lowest E#=1.1, it suffers from large positional errors. pix2pix performs worst on nearly all metrics, while Fast-DDPM yields noisy and inaccurate reconstructions.

The full SMO-Diff achieves optimal performance across all metrics, obtaining the highest PSNR (16.50±1.45 dB) and SSIM (0.61±0.03). Removing CMF causes moderate degradation: rod endpoint error increases to 10.3 px, missing screws rise to 1.1, and PSNR/SSIM drop to 15.91±1.15 dB / 0.57±0.02. Removing the Curve Encoder further impairs geometric consistency, with centerline error of 4.9 px and PSNR/SSIM of 15.87±1.21 dB / 0.58±0.02. Excluding the ROI Encoder leads to severe performance collapse: endpoint error surges to 55.2 px, missing screws reach 6.1, and PSNR/SSIM decline to 15.55±0.98 dB / 0.57±0.01.